

Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance

The report presents a survey of **Multi-Access Edge Computing (MEC) in 5G networks**, focusing on three important aspects: **security, dependability, and performance**. MEC brings computing and storage resources closer to end users, thereby reducing latency, network congestion, and dependence on centralized cloud services. It is particularly useful for applications such as **IoT, autonomous vehicles, industrial automation, and Ultra-Reliable Low-Latency Communication (URLLC)**.

The report explains the **ETSI MEC architecture**, which consists of MEC Hosts, virtualization infrastructure, MEC platforms, and management components such as the **MEC Orchestrator (MEO), MEC Platform Manager (MEPM), and Virtual Infrastructure Manager (VIM)**. MEC can be integrated with **5G networks, Network Function Virtualization (NFV), Software-Defined Networking (SDN), and network slicing** to provide flexible and efficient services.

Security is a major challenge because MEC nodes are distributed and may be deployed in locations with limited physical protection. The report discusses threats including eavesdropping, Man-in-the-Middle attacks, DoS/DDoS attacks, privacy violations, virtualization attacks, and unauthorized access. Encryption, authentication, access control, intrusion detection, secure communication, and AI/ML-based security mechanisms can help address these threats.

Dependability ensures that MEC services remain reliable and trustworthy. Important characteristics include availability, reliability, survivability, safety, and maintainability. Fault tolerance, redundancy, service migration, resource replication, and effective failure recovery are required to maintain service continuity, particularly in mobile and mission-critical applications.

Performance mainly concerns latency, throughput, jitter, packet loss, energy efficiency, and Quality of Service (QoS). Efficient resource allocation, MEC server placement, caching, congestion management, and service migration can improve performance. However, virtualization and security mechanisms may introduce additional overhead.

A key finding of the report is that **security, dependability, and performance are strongly interconnected**. Improving one aspect can negatively affect another; therefore, they must be considered jointly rather than independently. The report identifies **AI and machine learning, intelligent orchestration, MEC federation, and integrated monitoring** as important directions for future MEC and 6G networks.

In conclusion, MEC is a crucial technology for realizing the low-latency, reliable, and intelligent services expected from 5G and future 6G networks. Developing **secure, dependable, and high-performance MEC systems through integrated optimization and intelligent management** is essential for their successful deployment.

Reference:

G. Nencioni, R. G. Garroppo, and R. F. Olimid, “5G Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance,” *IEEE Access*, vol. 11, pp. 63496–63527, 2023



Received 11 May 2023, accepted 10 June 2023, date of publication 21 June 2023, date of current version 28 June 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3280334



TOPICAL REVIEW

5G Multi-Access Edge Computing: A Survey on Security, Dependability, and Performance

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This work was supported in part by the Norwegian Research Council through the Management and Orchestration for Data and Network Integration (5G-MODaNet) Project under Grant 308909, and in part by the Italian Ministry of Education and Research (MIUR) in the framework of the Future-Oriented Research Laboratory (FoReLab) Project (Departments of Excellence).

ABSTRACT The Fifth Generation (5G) of mobile networks offers new and advanced services with stricter requirements. Multi-access Edge Computing (MEC) is a key technology that enables these new services by deploying multiple devices with computing and storage capabilities at the edge of the network, close to end-users. MEC enhances network efficiency by reducing latency, enabling real-time awareness of the local environment, allowing cloud offloading, and reducing traffic congestion. New mission-critical applications require high security and dependability, which are rarely addressed alongside performance. This survey paper fills this gap by presenting 5G MEC's three aspects: security, dependability, and performance. The paper provides an overview of MEC, introduces taxonomy, state-of-the-art, and challenges related to each aspect. Finally, the paper presents the challenges of jointly addressing these three aspects.

INDEX TERMS 5G, MEC, security, dependability, performance.